

9 **(short)** A fidget spinner consists of four identical annular weights held in a triangular configuration by a light plastic casing. Each weight has mass m , outer radius a and inner radius b . The weights are separated by a distance $2b$ such that the centre of a weight in the spinner's arm is a distance $2(a+b)$ from the spinner's centre as shown in Fig. 9.

(a) Find the moment of inertia of the fidget spinner about an axis through its centre and perpendicular to the plane of the masses. [6]

(b) The angular velocity of the spinner about the central axis is increased linearly as a function of time t such that $\dot{\theta} = ct$, where c is a constant. Find the total force $F(t)$ on a single weight in an arm of the spinner, expressing your answer as a vector using polar coordinates. [4]

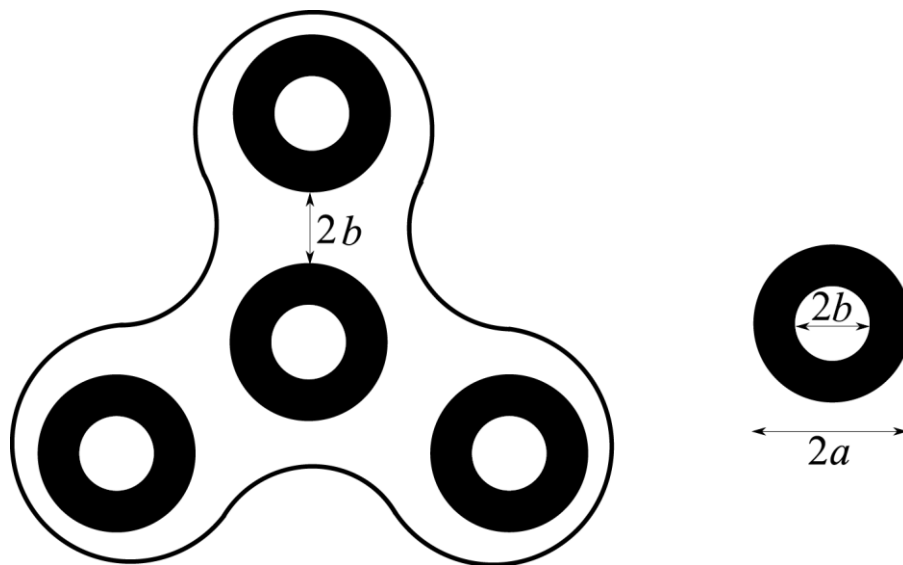


Fig. 9